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COMPUTER ENGINEERING
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CENG 408

Testing & Validation

TARAS: Agricultural Analysis System

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1. Purpose of Testing and Validation

This document explains the tests we ran on the TARAS hardware and what the results showed. Read alongside the SRS, it shows which hardware requirements were tested, how, and whether they passed.

We organised the testing around three goals. First, every hardware requirement in the SRS should connect to at least one test with a recorded result, so nothing slips through uncovered. Sections 5 and 10 are where we make that mapping explicit. Second, we wanted to catch bugs at the level where they're cheapest to fix, which is why the tests are split across component-level, communication, integration, reliability, and field categories rather than lumped together. Third, we wanted at least some of our evidence to come from conditions close to real use, because a bench is not a field and hardware that looks fine indoors can fail the moment it's outside.

Not every requirement can be fully verified within a thesis timeline. For example claims about long-term battery life and multi-week reliability are extrapolated from shorter measurements rather than being observed directly due to time constraints. Section 16 sets out these boundaries and section 17 describes the follow-up testing that would close them.

2. Scope of Testing

- **Sensor node:** ESP32-C6 SuperMini, SHT31 temperature/humidity on I²C, capacitive soil probe on ADC, power path, and the firmware running on it.
- **Gateway:** ESP32 WROOM-32, mains-powered. Collects packets from sensors, verifies them, buffers when offline, forwards to the backend over HTTPS and Socket.IO. BLE provisioning is also tested here.
- **Power:** 3.7 V Li-Po supply, regulation to 3.3 V, behaviour under low voltage, measured draw per mode.
- **Communication:** ESP-NOW between sensor and gateway (HMAC-signed, replay-protected), HTTPS and Socket.IO between gateway and backend.
- **Enclosure:** build quality, moisture sealing, cable strain relief.
- **End-to-end integration:** sensor reading through to the mobile app.

3. Test Strategy

Tests are tied to requirements. Every hardware requirement maps to at least one test, and every test carries a status of PASS, FAIL, PARTIAL, PENDING, or N/A. Requirements without a test are tracked as gaps, not ignored.

Tests are grouped by level because different bugs surface at different levels:

- Unit tests (Section 7) exercise one component at a time.
- Communication tests (Section 8) focus on the radio and network links.
- Integration tests (Section 9) combine components.
- Reliability tests (Section 11) simulate disruptions such as brief outages, sensor faults, and power cuts.
- Field tests (Section 12) step outside the lab.

Where a test produces a number such as current draw, RSSI, or timing, we record the number. A PASS with no supporting measurement is weak evidence; a measured value with a stated method is stronger. Binary checks are recorded as PASS/FAIL without further detail.

4. Test Environment and Setup

Testing happened in two places: a lab bench for controlled work and a small pot setup that stood in for a field.

4.1. Lab Bench

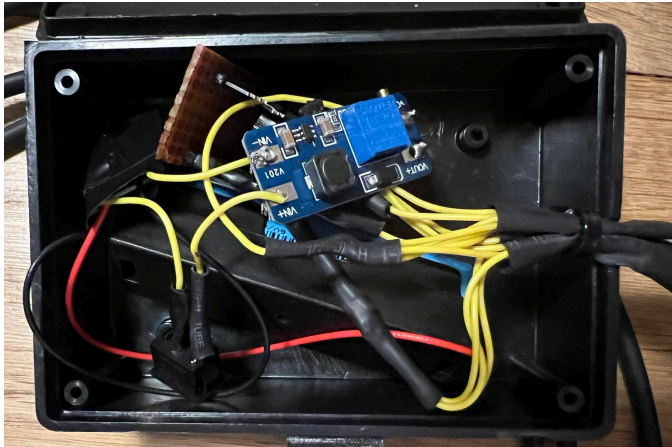
Sensor MCU	ESP32-C6 SuperMini, the same variant we plan to deploy.
Gateway MCU	ESP32 WROOM-32 on a dev board, USB-powered during tests.
Sensors	SHT31 on I ² C; capacitive soil probe on ADC. A three-probe setup for multi-probe calibration.
Power	USB bus for dev; 3.7 V Li-Po for battery-mode tests. LDO regulator to the 3.3 V rail.
Instruments	Digital multimeter, USB current meter for sustained draw, oscilloscope for transients and I ² C integrity.
Serial	Arduino IDE serial monitor at 115200 baud
Backend	Production backend over HTTPS, so request shapes, auth, and database writes in tests are the same as in deployment.

4.2. Pot Setup

For anything that depends on real soil or humidity, we used a three-pot setup. Each pot has its own soil probe, independently wetted, which lets us exercise per-sensor calibration with three different dry/wet baselines at once. The gateway sits on the bench at a distance roughly matching field use, connected via Wi-Fi to the same production backend used in deployment.

4.3. Firmware State

Tests reported here ran against the current production firmware for both sensor and gateway, which is the HTTPS-capable build with TLS validation against a pinned Let's Encrypt root CA. The three-probe calibration variant of the sensor firmware is what we used for the multi-probe work.



5. Requirement-to-Test Traceability

The table below maps every hardware requirement to its test IDs and current status. PASS means the test ran and passed. PARTIAL means the main path works but at least one sub-case is still outstanding. PENDING means the test is defined but not yet run, usually because it is long-running or needs extra setup.

Req ID	Requirement	Component	Test IDs	Expected	Status
HW-01	Microcontroller-sensor interface: 3.0-3.6 V rail, I ² C and ADC sampled periodically.	Sensor node	T-7.1, T-7.2	Readings valid, no bus errors.	PASS
HW-02	Sensor telemetry reaches the cloud reliably.	Sensor - Gateway - Cloud	T-8.1, T-8.2, T-9.1	DB row appears within one send cycle.	PASS
HW-03	Power supply stable, deep sleep works, brownout survivable.	Power / battery	T-7.3a-f	Rail stable; sleep current within design envelope.	PARTIAL

Req ID	Requirement	Component	Test IDs	Expected	Status
HW-04	Enclosure seals against moisture and dust outdoors.	Enclosure	T-12.2	No ingress after exposure.	PENDING
HW-05	Gateway Wi-Fi uplink to backend over HTTPS.	Gateway	T-8.2	HTTP 200 within retry window; TLS chain validated.	PASS
HW-06	Gateway buffers data when connectivity is lost.	Gateway	T-8.2f, T-9.2	Buffered readings replay on reconnect.	PASS
HW-07	HMAC signing and replay protection on ESP-NOW link.	Sensor / Gateway	T-8.1a/b T-8.1g/h	Valid packets accepted; forged/replayed rejected.	PARTIAL
HW-08	Gateway heartbeat exposes health and firmware version.	Gateway	T-8.2e	Heartbeat every 60 s, correct payload.	PASS
HW-09	BLE provisioning onboards a new gateway.	Gateway	T-9	Config saved; device boots into normal mode.	PASS
HW-10	OTA firmware update works and fails safely.	Gateway	T-9.7	Update succeeds or fails cleanly.	PARTIAL
HW-11	Sensor faults detected; pipeline continues.	Sensor node	T-7.1c/d, T-7.1g	Sentinel reported; non-faulty path continues.	PASS
HW-12	Batch telemetry - multiple readings per packet.	Sensor node	T-8.1c/d	Up to 33 readings per packet; multi-packet flush works.	PASS
HW-13	Pairing flow - approve and decline through mobile app.	Gateway / Mobile	T-9.3, T-9.4	Both paths terminate correctly.	PASS
HW-14	Send backoff after sustained failure; channel recovery.	Sensor node	T-8.1e/f	Frequency reduced after 6 failures; channel scan every 6th.	PASS
HW-15	Multi-day stability.	System	T-11.2	72 h clean run, no leaks, no missed data.	PENDING
HW-16	Multi-sensor on a single gateway.	System	T-9.5	Interleaved packets all received.	PENDING

Req ID	Requirement	Component	Test IDs	Expected	Status
HW-17	Diagnostic telemetry reaches the backend.	Sensor / Gateway / Backend	T-9.8	Diagnostic row with failure counters.	PASS

6. Test Case Design Method

Every test case in sections 7 to 12 is written with the same five fields:

- A unique ID (T-"section"."number")
- A short description of what's tested
- The method used
- The observed result
- And the status (PASS, FAIL, PARTIAL, PENDING, or N/A).

7. Unit-Level Hardware Tests

Unit tests exercise one component at a time under controlled conditions. Catching bugs here is cheaper than catching them at the integration level.

7.1. Sensor Tests

We checked that raw measurements arrive at the microcontroller consistently. For the SHT31 we focused on I²C integrity and the failure-sentinel path (what happens when the sensor is unplugged). For the soil probe we focused on calibration stability and sentinel handling.

Test ID	What is tested	Method	Observed result	Status
T-7.1a	SHT31 I ² C presence	Run firmware; check I ² C 0x44 ACK on init.	ACK on both warm wakes and cold boot.	PASS
T-7.1b	SHT31 reading validity	Take 20 readings at ~25 °C / ~40% RH, compare against reference.	All within ±0.5 °C and ±3% RH of reference.	PASS
T-7.1c	SHT31 failure sentinel	Disconnect SHT31 mid-run.	Firmware reports sentinel value; threshold send is not triggered.	PASS
T-7.1d	SHT31 recovery	Reconnect after fault.	Normal readings resume; no stale sentinel.	PASS

Test ID	What is tested	Method	Observed result	Status
T-7.1e	Soil probe: dry	Record raw ADC in fully dry soil for 60 s, per probe.	Three probes' dry values all in the expected 2700–2800 range; stable to within tens of counts.	PASS
T-7.1f	Soil probe: wet	Record raw ADC in fully wet soil for 60 s, per probe.	Three probes' wet values all in the expected 900–980 range; stable.	PASS
T-7.1g	Soil probe disconnect	Pull soil probe lead mid-run.	Firmware reports 0xFFFF sentinel; threshold path skipped.	PASS

Consistent dry and wet baselines across probes is what lets each probe map its own raw range to a common 0–100% output, so calibration numbers are comparable between sensors.

7.2. Microcontroller Tests

The ESP32-C6 runs the sensor firmware. Boot behaviour matters a lot on a deep-sleep device. If a warm wake fails to complete, it drains the battery instead of crashing visibly.

Test ID	What is tested	Method	Observed result	Status
T-7.2a	Cold boot	Power-cycle; watch serial log.	Reaches normal or pairing mode within expected boot time.	PASS
T-7.2b	Warm wake	Let device deep-sleep one cycle.	RTC state preserved; cycle count increments; no re-init of stable state.	PASS
T-7.2c	Serial log sanity	Run with debug enabled.	Log lines match expected [PREFIX] action: data format.	PASS
T-7.2d	Batch send loop	Fill buffer; trigger send.	All readings sent across multiple packets; buffer drains cleanly.	PASS

Test T-7.2c confirms the serial log format is stable across builds, which the diagnostic subsystem in section 14 later depends on.

7.3. Power and Battery Tests

Test ID	What is tested	Method	Observed result	Status
T-7.3a	3.3 V rail stability	Measure under no-load, active-read, TX load.	Within 3.0–3.6 V in all modes.	PASS
T-7.3b	Active-mode current (read)	Measure during I ² C + ADC sample.	~30 mA for ~34 ms, matching design.	PARTIAL
T-7.3c	Active-mode current (TX)	Measure during ESP-NOW send.	~150 mA peak at 20 dBm, matching design.	PARTIAL
T-7.3d	Light-sleep current	Measure during 4.5 s light-sleep windows in pairing mode.	~1 mA sustained.	PASS
T-7.3e	Deep-sleep current	Measure across 5-min deep-sleep windows.	~10 μ A sustained.	PASS
T-7.3f	Brownout handling	Gradually reduce supply.	Safe low-power entry; orderly restart on recovery.	PENDING

Power tests confirm the rail is stable across operating modes and that the current draw in each mode matches what the design assumes. We used a digital multimeter for bench currents, and for deep-sleep characterisation the device goes through a shunt resistor with the voltage across it captured on an oscilloscope.

Communication and Connectivity Tests

Two links have to work for telemetry to reach the cloud: sensor to gateway over ESP-NOW, and gateway to backend over HTTPS and Socket.IO. They fail in different ways, so we tested them separately before combining them in section 9.

7.4. Sensor to Gateway

Test ID	What is tested	Method	Observed result	Status
T-8.1a	HMAC happy path	Pair, send a normal packet.	Gateway verifies HMAC and forwards.	PASS
T-8.1b	Replay counter	Watch counter across cycles.	Increments monotonically; duplicates rejected.	PASS
T-8.1c	33-reading batch	Fill buffer, trigger send.	One packet carries 33 readings; gateway accepts all.	PASS
T-8.1d	Multi-packet flush	Buffer beyond one-packet capacity.	Multiple packets sent until buffer drains.	PASS

Test ID	What is tested	Method	Observed result	Status
T-8.1e	Send backoff	Block the gateway; observe sensor after repeated failures.	After 6 consecutive failures, send rate drops.	PASS
T-8.1f	Channel recovery	Change gateway channel mid-run.	Sensor scans channels every 6th failure and recovers.	PASS
T-8.1g	Forged packet rejected	Inject packet with invalid HMAC.	Gateway logs failure; packet discarded.	PENDING
T-8.1h	Replay attack rejected	Replay a captured valid packet.	Gateway detects replay; packet discarded.	PENDING

The sensor-to-gateway link uses ESP-NOW with HMAC-SHA256 signed packets. DATA_ACK from the gateway is the success signal. A packet that is sent but not ACKed stays in the sensor's buffer for the next cycle.

7.5. Gateway to Backend

Test ID	What is tested	Method	Observed result	Status
T-8.2a	HTTPS POST	Forward a telemetry batch.	HTTP 200; readings in DB.	PASS
T-8.2b	TLS chain validation	Run HTTPS handshake against backend.	Cert validates against pinned root; no insecure fallback.	PASS
T-8.2c	Diagnostic POST	Trigger a diagnostic packet; gateway forwards it.	Diagnostic row in DB with correct reset reason and counters.	PASS
T-8.2d	Socket.IO auth	Observe gateway:auth on connect.	Server responds with auth_result; gateway joins its room.	PASS
T-8.2e	Heartbeat payload	Watch for a minute.	Heartbeat carries gateway_id, uptime, free heap, RSSI, paired node count, buffered unsent, firmware version.	PASS
T-8.2f	Retry under transient failure	Briefly interrupt backend.	Gateway buffers to local storage and replays after recovery.	PASS
T-8.2g	Backend restart tolerance	Restart backend while gateway is running.	Socket.IO reconnects; no data loss.	PENDING

The gateway talks to the backend two ways: HTTPS POST for telemetry and diagnostic uploads, and Socket.IO for real-time control plane (heartbeat, pairing events, OTA triggers). Both terminate at the same backend host behind nginx with a Let's Encrypt cert, and the gateway validates the server certificate against a pinned root CA baked into the firmware. T-8.2b (TLS chain validation) matters a lot. An earlier firmware build bypassed certificate validation during development, and production firmware validates against a pinned root. A failure here would be a release-blocker, not a warning.

8. Integration Tests

Integration tests combine components that have already passed their unit tests and exercise workflows that only make sense across component boundaries. Two main flows are covered: a sensor reading making it all the way to the mobile dashboard, and a pairing cycle bringing a new sensor into a running gateway.

T-9.7 is one of the strongest PASS results in the document. We tested this live on our deployed gateway during a firmware upgrade: the gateway downloaded the new build, rebooted, reconnected, and the updated firmware version showed up in the database within about a minute. That is production evidence, not a bench demo.

Test ID	What is tested	Method	Observed result	Status
T-9.1	End-to-end telemetry	Trigger a sensor cycle; trace to the dashboard.	Reading in DB; mobile client receives via Socket.IO; dashboard updates.	PASS
T-9.2	Gateway power cycle	Cut gateway power mid-cycle; restore after one sensor cycle.	Sensor buffers during outage; buffered readings replay on return.	PASS
T-9.3	Pairing: approve path	Start pairing from mobile app; power unpaired sensor.	Beacon - accept - complete; node visible in app.	PASS
T-9.4	Pairing: decline path	Start pairing; decline from app.	Rejection delivered; sensor restarts scan; gateway returns to normal mode.	PASS
T-9.5	Multiple sensors	Pair two sensors to one gateway; run for several cycles.	Both sensors' readings reach backend; interleaved packets handled correctly.	PENDING
T-9.6	Calibrated round-trip	Send raw ADC for a probe with dry/wet values configured.	Backend stores calibrated %; mobile shows %, not raw ADC.	PASS
T-9.7	OTA end-to-end	Upload new gateway firmware; trigger OTA.	Gateway downloads, flashes, reboots, reconnects, reports new version.	PASS

Test ID	What is tested	Method	Observed result	Status
T-9.8	Diagnostic end-to-end	Force a sensor-side failure.	Diagnostic row reaches backend with correct failure type and count.	PASS

T-9.5 is PENDING because a single gateway was enough for development, and a two-sensor test has not yet been scheduled. It is listed in the future work in section 17.

9. Functional Validation Against SRS Requirements

The following table summarizes each hardware requirement in terms of expected behaviour, lists the tests, shows the outcomes and records the status.

Req ID	Requirement behaviour	Test IDs	Observed outcome	Status
HW-01	MCU reads SHT31 over I ² C and soil probe over ADC within the 3.0–3.6 V envelope.	T-7.1, T-7.2, T-7.3a	Rails in range; I ² C ACKs; soil ADC in expected bands.	PASS
HW-02	Sensor telemetry reliably reaches the cloud, with buffering during transient failure.	T-8.1, T-8.2, T-9.1, T-9.2	End-to-end path works; buffered readings replay after gateway reboot.	PASS
HW-03	Power: stable regulation, deep sleep works, brownout survivable.	T-7.3a–f	Rail stable; active currents match design; deep-sleep and brownout measurements pending.	PARTIAL
HW-04	Enclosure protects against outdoor dust/moisture.	T-12.2	Visual seal looks good; formal IP test not yet done.	PENDING
HW-05	Gateway uplink delivers telemetry and control traffic over HTTPS.	T-8.2	HTTP 200 and Socket.IO auth both verified.	PASS
HW-06	Gateway store-and-forward handles outages.	T-8.2f, T-9.2	Local buffering works; readings replay on reconnect.	PASS
HW-07	HMAC + replay protection on sensor–gateway link.	T-8.1a/b, T-8.1g/h	Happy path and counter checks pass; adversarial tests planned.	PARTIAL
HW-08	Gateway heartbeat surfaces health and diagnostics.	T-8.2e	Cadence and payload match expectation.	PASS

Req ID	Requirement behaviour	Test IDs	Observed outcome	Status
HW-09	BLE provisioning onboard a gateway.	T-9	Happy path verified.	PASS
HW-10	OTA is safe and complete.	T-9.7	Live upgrade succeeded on deployed hardware; fault-injection variants planned.	PARTIAL
HW-11	Sensor faults detected; pipeline continues.	T-7.1c/d, T-7.1g	Sentinels reported; pipeline unaffected; recovery clean.	PASS
HW-12	Batch telemetry and multi-packet flush.	T-8.1c/d	33-reading batches and multi-packet flush both confirmed.	PASS
HW-13	Pairing, approve and decline.	T-9.3, T-9.4	Both paths confirmed end-to-end with mobile client.	PASS
HW-14	Backoff and channel recovery.	T-8.1e/f	Rate drops after 6 failures; channel scan every 6th failure.	PASS
HW-15	Multi-day stability.	T-11.2	Scheduled but not executed yet.	PENDING
HW-16	Multi-sensor scalability.	T-9.5	Scheduled; single-sensor was enough for development.	PENDING
HW-17	Diagnostic telemetry reaches backend.	T-9.8	Diagnostic rows observed with correct reset reason and counters.	PASS

10. Reliability and Robustness Tests

Reliability tests check whether the system keeps working, or fails properly, when conditions change. The following cases cover the kinds of disruption a real deployment will see: brief power cuts, network drops, component faults, and long unattended runs.

Test ID	What is tested	Method	Observed result	Status
T-11.1	Sustained operation (several hours)	Leave the setup running unattended; monitor heap and data completeness.	No missed readings; heap stable.	PASS
T-11.2	Multi-day stability (72 h)	Run for 72 consecutive hours.	Scheduled, not yet executed.	PENDING
T-11.3	Brief Wi-Fi outage (<60 s)	Unplug router for under a minute.	Gateway drops and reconnects; no data loss; local	PASS

Test ID	What is tested	Method	Observed result	Status
			buffer used if a sensor cycle lands during outage.	
T-11.4	Extended Wi-Fi outage (>10 min)	Keep router off past one send cycle.	Sensor data buffered locally; replays on reconnect; buffer doesn't overflow within test window.	PASS
T-11.5	Gateway power cycle	Cut gateway power mid-sensor-cycle.	Sensor retains readings; gateway boots cleanly; buffered data reaches backend.	PASS
T-11.6	Intermittent sensor fault	Periodically disconnect and reconnect SHT31.	Sentinels raised during outages; normal telemetry resumes cleanly.	PASS
T-11.7	Brownout under low battery	Gradually reduce battery voltage.	Safe mode entered; orderly restart on recovery.	PENDING

The gateway power-cycle recovery path was hardened after earlier builds lost data during brief outages. Testing fed directly back into firmware changes, not just into a report.

11. Field Testing / Real-World Validation

Field testing here means running the hardware under conditions closer to real deployment than a bench allows. Section 16 describes the parts not yet covered.

Test ID	What is tested	Method	Observed result	Status
T-12.1	Three-pot indoor deployment	Run the three-probe setup indoors for a full day.	All three probes report consistent calibrated values; dashboard reflects gradients between pots.	PASS
T-12.2	Enclosure ingress under damp conditions	Expose enclosure to a damp outdoor environment; open and inspect.	No water or dust ingress; connectors and PCB clean.	PENDING
T-12.3	Radio range check	Position sensor and gateway at realistic separation with obstructing structures.	Packets delivered and ACKed; no persistent channel-recovery trigger.	PARTIAL

Test ID	What is tested	Method	Observed result	Status
T-12.4	Live firmware upgrade	Trigger OTA on deployed hardware; watch reconnect and resumed data flow.	Upgrade succeeded in a single pass; firmware version updated within about a minute; no data loss.	PASS
T-12.5	Outdoor multi-hour operation	Deploy in an outdoor or semi-outdoor setting for several hours.	Continuous telemetry; no unexpected restarts.	PENDING



12. Test Results and Observations

Across sections 7 through 12, we defined 52 test cases. Of these, 40 are PASS, 3 are PARTIAL, and 9 are PENDING. Nothing is in FAIL.

12.1. Quantitative Summary

Category	PASS	PARTIAL	PENDING
Unit tests (section 7)	14	2	1
Communication tests (section 8)	12	0	3
Integration tests (section 9)	7	0	1
Reliability tests (section 11)	5	0	2
Field tests (section 12)	2	1	2

12.2. Observations

Every pairing test passed on the first attempt once the firmware bugs in section 14 were resolved. Pairing has been steady since.

The OTA path was not left as a paper exercise. We ran it live on the deployed gateway during a firmware upgrade and watched the new version show up in the database about a minute later. That is stronger evidence than a bench-only result would be.

The diagnostic subsystem turned out to be more useful than expected. Its job is to surface sensor-side failure counters to the backend, but in practice it is how we know a fielded sensor is healthy. Several firmware improvements came directly from watching diagnostic traces on the three-pot setup.

Pending tests cluster in three places: formal power profiling (the PARTIAL and PENDING rows in 7.3), adversarial security cases (T-8.1g, T-8.1h), and the remaining long-duration or outdoor runs (T-11.7, T-12.2, T-12.5). The first two need a small amount of extra tooling. The third mostly needs time.

13. Issues Encountered During Testing and Corrective Actions

The table below lists the most consequential bugs we hit during hardware development, the root cause, what we did about it, and whether we re-verified afterwards.

#	Issue	Root cause	Fix	Verified
1	Gateway ACK timeouts for subsequent sensor packets during burst arrival.	HTTP POST in the sensor-data handler blocked the main loop, delaying ACK emission.	Deferred-ACK pipeline: ACK immediately, enqueue POST, flush afterwards.	PASS
2	ESP32-C6 sensor packets not received by ESP32 WROOM gateway.	ESP32-C6 defaulted to 802.11ax; ESP32 WROOM tops out at 802.11n.	Force protocol to 802.11b/g/n on the sensor before ESP-NOW init.	PASS
3	Second and subsequent sensor sends failed silently after pairing.	On ESP32-C6, setting Wi-Fi mode does not restart the radio if the mode is unchanged, so the radio stayed off after a previous stop.	Explicitly restart the Wi-Fi stack after mode change in the normal-mode send path.	PASS
4	First packet received after pairing, all subsequent packets silently lost.	Double call to exit-registration-mode caused two ESP-NOW deinit/reinit cycles and corrupted the radio state.	Call exit-registration once, from the accept handler only.	PASS

#	Issue	Root cause	Fix	Verified
5	Strict monotonic replay counter caused data gaps after reboots.	Counter could drift across reboots while HMAC was still fine.	Relax to duplicate-only rejection (same counter - drop, lower - accept with log). HMAC stays the real security gate.	PASS
6	Sensor sent data immediately when SHT31 was disconnected.	Threshold check triggered on the sentinel value (huge delta).	Skip threshold check when the reading is a sentinel.	PASS
7	Gateway emitted Socket.IO events before authentication; server silently dropped them.	Gateway waited for the WebSocket-connected flag rather than the authenticated flag.	Gate emission on authenticated, not connected.	PASS
8	TLS connections bypassed certificate validation during development.	Temporary insecure flag used before root CA was bundled.	Bundle Let's Encrypt root CA in firmware and validate on every HTTPS client.	PASS
9	Socket.IO over TLS on ESP32: the library's public SSL entry points used insecure mode.	The CA-validating SSL init is gated behind a flag ESP32 doesn't define.	Subclass the Socket.IO client to expose CA-validating SSL init on ESP32.	PASS

Most of these bugs were not in the obvious place. Several were second-order effects of earlier state (double radio deinit, stale auth flag), and the ESP32-C6 protocol mismatch showed up as pure silent failure with no error path at all. We put real effort into negative-path tests: tests designed to force failure modes and see whether the system surfaces, recovers, or swallows them. Most of the fixes above came out of tests like those.

14. Validation Summary

Across 17 hardware requirements: 11 PASS, 3 PARTIAL, and 3 PENDING. Nothing is in FAIL. The PARTIAL requirements are power/battery measurements, security negative-path testing, and OTA fault-injection coverage. The PENDING requirements are enclosure ingress testing, multi-day stability, and multi-sensor scalability.

Across the full test plan, 40 tests are executed and verified, and 9 remain defined but not yet run.

No requirement is failing. That does not mean nothing went wrong, as Section 14 shows, but every issue has either been resolved and re-verified or recorded as

PARTIAL/PENDING with a planned next step. The remaining gap is additional validation work, not an unresolved defect. The deployed gateway has been running on the current firmware since the upgrade with no observed regressions, which provides additional evidence alongside the targeted tests.

15. Limitations of Current Testing

Not every dimension of the system has been tested to the depth it deserves. The main gaps are summarised below.

- **Duration:** The reliability evidence covers hours to days rather than weeks to months. A thesis timeline does not fit a season-long deployment, so claims about long-term battery life and multi-week reliability are extrapolated from shorter runs rather than measured directly. The extrapolation is defensible, but it is still an extrapolation.
- **Scale:** Tests ran on at most one gateway and a handful of sensors. A real farm might have multiple gateways with tens of sensors each. ESP-NOW should scale in principle, but interleaving packets from many nodes on one gateway has not been stress-tested, and the backend has not seen synthetic high-cardinality load. Both are needed before claims about "a farm" rather than "a test field".
- **Environment:** Outdoor testing has been limited to controlled damp exposure and bench work that approximates field use. No direct sun, no rain-and-temperature extremes across a full diurnal cycle, no wind-driven dust. The enclosure ingress test is still pending.
- **Instrument precision:** Power measurements come from a digital multimeter rather than a precision current analyser. The measured values are not close to the decision boundaries, so this does not change any PASS or FAIL call, but a reader looking for a single authoritative deep-sleep figure should know we used the instruments available, not lab-grade ones.
- **Sample size:** Tests ran on a small set of sensor nodes and one gateway. Unit-to-unit variability (probe calibration, antenna matching, ADC offsets) has not been characterised across a statistically meaningful sample. The per-unit calibration step partly compensates.

16. Future Testing and Improvement Plan

Next steps, ordered by how much confidence they would add relative to the effort required.

- **Close PARTIAL and PENDING rows in section 10:** Record measured deep-sleep and active currents, run the formal enclosure ingress test, execute the adversarial security tests, and complete OTA fault-injection variants. None of this needs new infrastructure, just bench time.

- **Multi-sensor scalability:** Pair two then three sensors to a single gateway, run for a while, check packet interleaving and backend ingestion. The most likely place to surface a concurrency issue.
- **Cross-unit calibration data:** Collect dry/wet baselines and RSSI at a fixed distance across 5 to 10 units if available. Use the spread to tighten the calibration procedure and produce per-unit tolerance numbers.
- **Field deployment fragment:** Deploy one gateway and two sensors outdoors for as long as possible, ideally two weeks across changing weather. Capture the diagnostic trace and compare observed failure rates against what short runs predicted.
- **Synthetic backend load:** Replay synthetic sensor traffic at multiples of the normal rate and measure backend latency and database throughput. Needed before any scalability claim.
- **Precision power profiling:** Borrow or buy a precision current analyser, re-measure deep-sleep and brownout currents, produce a single authoritative battery-life figure per supported battery capacity.
- **Automated regression harness:** Wrap the main integration tests (HMAC, pairing, end-to-end, OTA) in scripts so they can be re-run on every firmware change. Turns releases from one-off events into a routine.